MINING
INDUSTRY REVIEW

Rowan Das
October 2026

Opening Statement

Mining is the science, technique and business of finding mineral deposits and getting them out of the ground at a profit. Metallurgy, the study of metals and their properties, covers what happens next: turning rock into metal someone will pay for. It still sits underneath most of the economy, from the copper in a power line to the iron in a steel beam. Miners are price takers. They sell into global markets they can't control, so earnings rise and fall with the commodity and the economy. In 2025 the 40 largest miners grew revenue only 3.3% to US$909bn, but EBITDA jumped 23% to US$248bn as gold and copper prices climbed. That sums up the business: volumes barely move, prices do. It is also important to mention that resources are finite hence firms intuitively have a finite life-span.

Key Terms

Exploration. The search for new deposits, starting with a desktop study of existing maps, surveys, old drill results and satellite data before anyone goes into the field.

Ore. Rock that holds enough valuable mineral to be worth mining and processing at current prices. The same rock can be ore at US$14,000 copper and waste at US$6,000.

Resource vs. reserve. A resource is mineralization with reasonable prospects of being mined at a profit; a reserve is the part a study shows can actually be mined economically (page 2).

Industry Characteristics

Cyclical price takers. Revenue follows the commodity and economic cycle, so value depends on where we are in the cycle and how volatile prices, earnings and cash flow are expected to be.

High fixed costs and debt. A mine costs about the same to run whether prices are high or low, so small price moves swing operating income. Many miners borrow to build, so equity values and debt ratios swing too.

Long lead times. Discovery to production averaged 15.7 years across 127 mines (S&P Global), with financing, permitting, geology, metallurgy and economic risk along the way.

Country risk. Deposits are where they are, often in places with political, economic or geographic risk: royalty hikes, export bans or no roads and power.

Scarcity. Every tonne mined is a tonne gone, so without new exploration production would fall away quickly.

Key Stats

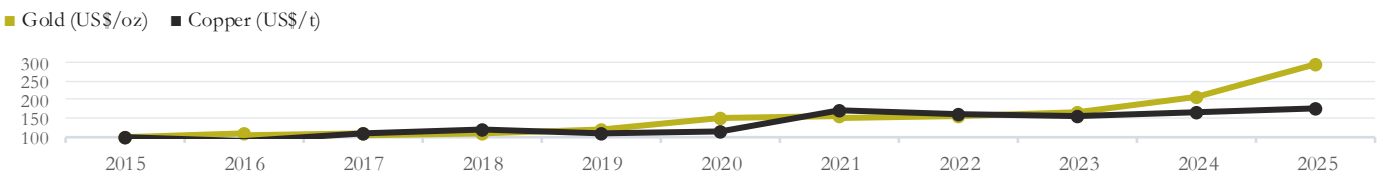
Top 40 Miners, 2025	
Revenue	$909bn
Revenue growth	+3.3%
EBITDA (growth)	$248bn (+23%)
EBITDA margin	27%
Net profit	$120bn
Market cap	**$1,956bn**
Prices	
Gold, 2025 average	$3,432/oz
Gold record, 28-Jan-26	$5,590/oz
Gold, 15-Oct-26	$4,176/oz
Copper, 15-Oct-26	**~$14,400/t**
Discovery to Production	
Average, 127 mines	15.7 yrs
Mines started 2020-24	**17.8 yrs**



Metal Groups and What They Are Used For

Group	Examples	Main uses	What drives the price
Precious	Gold, silver, platinum group	Investment and central bank reserves, jewellery, electronics, catalytic converters	Real interest rates, the U.S. dollar, central bank buying, fear
Base	Copper, nickel, zinc, lead, aluminium	Wire and power grids, construction, EVs and batteries, galvanizing steel	Industrial output, construction and China
Bulk	Iron ore, metallurgical and thermal coal	Steelmaking and power	Chinese steel output and infrastructure
Minor / critical	Cobalt, lithium, graphite, silicon, rare earths	Batteries, magnets, chips and solar	EV and tech demand, policy, Chinese supply

Gold and Copper, Annual Average Price (Indexed, 2015 = 100)



MINING
MINE LIFE CYCLE

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How a Deposit Becomes a Mine

A mine gets built in stages, and each stage either adds confidence or kills the project. Most deposits never become mines. Value is created as risk comes out, which is why a junior's share price can double or halve on a single drill result or study.

The Study Pipeline

Before a company commits to building, it works through a series of studies, each more detailed and expensive than the last.

Scoping study. A first look at whether the deposit could work. In Canada this is a preliminary economic assessment (PEA), and it can lean on inferred resources.

Pre-feasibility study. Compares mining and processing options and picks one. It is the first stage at which reserves can be declared.

Feasibility study. Detailed engineering, costs and schedule for the chosen plan.

Bankable feasibility study. A feasibility study detailed enough that lenders will finance the build against it.

Resources and Reserves

Under Canada's NI 43-101 rules and the CIM definitions, resources are ranked by geological confidence. Inferred resources rest on limited drilling, indicated resources on enough drilling to estimate grade and tonnage reasonably well, and measured resources on close-spaced drilling that confirms the deposit is continuous. Before a production decision, resources have to be proven up: indicated resources can become probable reserves and measured resources can become proven reserves, once a study shows they can be mined at a profit. Inferred resources can't be converted at all.

Exploration Properties

An exploration property is one where no deposit has been found yet. It has no cash flow and often no assets beyond the claims and the data, so its value is the chance of a discovery. That is why exploration companies are valued on spending and comparable deals rather than a DCF (page 3). Exploration matters for the whole industry: the more a company mines, the less it has left, and without new discoveries production would cease quite quickly.

How a Mine Makes Money

Revenue = commodity price × annual volume. Volume depends on the rate of ore production, the grade of that ore and metallurgical recovery, the share of metal the plant actually gets out. Price is set by the market, so the levers a miner controls are tonnes, grade, recovery and cost.

Types of Mining

Surface (open pit). Digs from the top down and suits large, shallow, lower-grade deposits.

Underground. Reaches deeper, higher-grade ore through shafts and ramps. The deepest gold mines in South Africa work about 4 km down, still only the top of a crust that is roughly 7 km thick under the oceans and 35 km under the continents.

Placer. Uses water and gravity to separate heavy minerals like gold from loose sand and gravel.

In-situ. Pumps fluid into the deposit through wells and brings the dissolved metal back up, common for uranium.

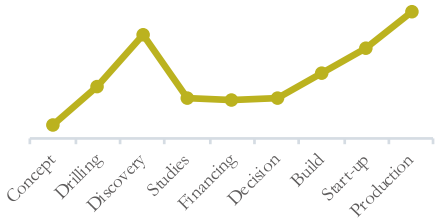
Dredging. Recovers ore from under water, such as river or offshore sands.

Resource to Reserve

Resource	Confidence*	Can become
Inferred	~10%+	Nothing yet
Indicated	~50%+	Probable reserve
Measured	~90%+	Proven reserve

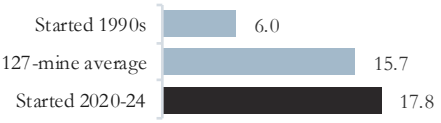
**Rough chance the mineralization is there. A rule of thumb, not part of the CIM definitions.*

The Life Cycle of a Mine (Share Price)



Speculation peaks at discovery, reality sets in through studies and financing, and the stock re-rates once it produces. The dip around the production decision is the classic entry point. Illustrative, after the Lassonde curve.

Discovery to Production (Years)



1990s figure is approximate (S&P: 2020-24 is nearly three times longer).



MINING VALUATION

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October 2026

There Is No Single Method

How you value a mining asset depends on how far along it is. An exploration property has no cash flow to discount, so it is valued on what has been spent on it and what similar ground has sold for. A producing mine is valued mostly on the cash it will generate over the life of its reserves. Canada's CIMVal standards group the methods into three approaches: income, market and cost.

Valuation Approaches by Property Stage

Approach	Method	Exploration	Development	Production
Income	Discounted cash flow	Not generally used	**Widely used**	**Widely used**
	Real options	Less widely used	Quite widely used	Quite widely used
	Monte Carlo analysis	Less widely used	Less widely used	Less widely used
	Probabilistic methods	Not widely used	Not widely used	Not widely used
Market	Comparable transactions	**Widely used**	**Widely used**	**Widely used**
	Option agreement terms	**Widely used**	**Widely used**	Quite widely used
	Value per unit area	**Widely used**	Not widely used	Not widely used
	Net metal value per unit of metal	*Widely used rule of thumb*		
	Market capitalization	*Mainly for single-property junior companies*		
	Gross in-situ metal value	*Not acceptable*		
Cost	Appraised value	Quite widely used	Not widely used	Not generally used
	Multiples	Quite widely used	Quite widely used	**Widely used**
	Geoscience factor	Not widely used	Not widely used	Not generally used

Valuation methods after Svetlana Baurens & University of Zurich

Income Approach: DCF and NAV

For a producing or near-production mine, analysts build a DCF over the life of the reserves and add up the projects into a net asset value (NAV). The DCF only works if the price deck is right, which is why the long-term trend matters more than today's price (see right).

Real Options

An undeveloped deposit works like a call option: the owner can pay the build cost to get the mine's value, but doesn't have to and can wait. The investment can't be reversed, the future reward is uncertain and it doesn't have to happen right away, which is exactly when options pricing beats a static DCF. Once a mine is running, the right to shut it or mothball it when prices fall works more like a put.

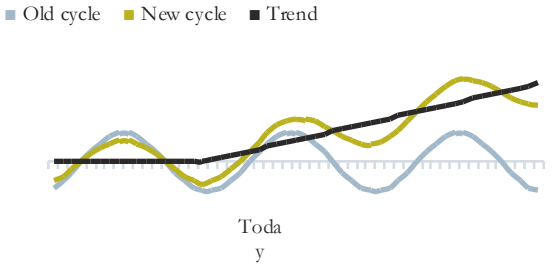
Market Approach

Comparable transactions give a benchmark, but only a rough one: exploration itself is cyclical, so deals done at the top of the cycle make every later deal look cheap. Common ratios are price to NAV (market cap or adjusted market cap, counting core mining assets only, over NAV) and equity value per unit of resource, such as EV per ounce of gold or per pound of copper.

Cost Approach

The appraised value method adds retained past expenditures (useful exploration already done) to warranted future costs (a reasonable exploration budget). It suits properties that are actively being explored and have no resource yet.

When the Cycle Changes



Earnings over time. Illustrative, after McKinsey & Company (2000).

Managing the Cycle

Fixating on one base year goes wrong in a predictable way: value off a peak year and you overvalue the company, value off a trough and you undervalue it. The inputs get skewed too, including debt ratios, reinvestment, profitability and even the risk-free rate.

The fix is to normalize: use a long-term price deck and mid-cycle margins so that, over a full cycle, high and low cash flows roughly cancel out. What really drives value is the long-term trend, which for metals is set by urbanization (page 4). The hard part is spotting when the trend itself shifts, as in the chart.

Source: CIMVal Standards and Guidelines for Valuation of Mineral Properties (CIM), as reproduced by Basinvest; McKinsey & Company (2000), 'When the cycle changes'; course notes.

MINING
KEY MACRO DRIVERS

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Urbanization Sets the Trend

When people move to cities they need housing, power lines, transit and water systems, and all of that is steel, copper and aluminium. That is why China drove the last metals supercycle. Its urban share rose from 36% in 2000 to 66% in 2025, and its share of world base metal demand rose with it: by 2009 China used 45-46% of the world's base metals and 46-49% of its steel, up from about 33% and 34% in 2008. Today it uses about 54% of the world's refined copper. India is now about 36% urban, roughly where China was in 2000, which is why it gets talked about as the next big source of demand.

Current Macro Trends

Prices, Not Volumes, Are Driving Profits

PwC's Top 40 grew revenue just 3.3% in 2025 while EBITDA rose 23%. Gold miners ran EBITDA margins near 71%, copper EBITDA rose 80% to US$13.3bn, and coal revenue fell 10.7%. Gold averaged US$3,432 in 2025, up 44%, peaked at US$5,590 in January 2026 and has since fallen about 25%. Copper hit a record near US$14,900/t in 2026.

Productivity Is Stuck

Output per worker has fallen since 2020 and capital productivity was flat in 2025, even as capex rose to US$95bn. Mining scored lowest of any sector on PwC's AI fitness index, and 40% of mining CEOs say their technology is underdelivering, so AI hasn't shown up in the numbers yet.

Trade Restrictions

China produces more than half of 18 key minerals and dominates processing. It began licensing seven rare earths in April 2025 and widened controls to graphite anode material and more rare earths in October 2025, then paused the October rules until November 10, 2026. Indonesia, the world's biggest nickel supplier, has banned raw ore exports since 2020.

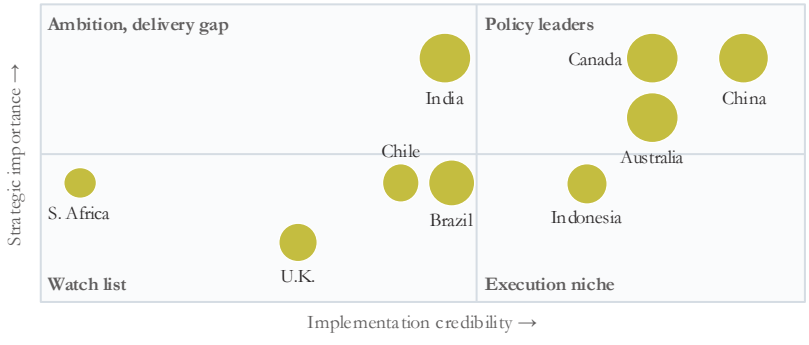
Satellites, Data and AI in Exploration

Satellite imagery and data tools now produce prospectivity heat maps, recommended drill targets and 3D models of a deposit. It is not a big trend yet, but it could cut the cost and time of finding new deposits.

Infrastructure, ESG and Policy

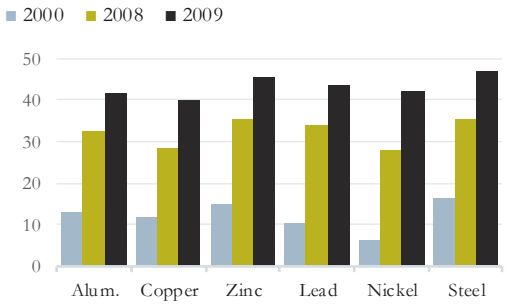
A mine is worthless without power lines, roads, rail and a port, so mining spend is tied to infrastructure. ESG matters most in Canada, where Indigenous consultation and permitting decide timelines. Policy is now a deciding factor, with governments treating critical minerals as a security issue (chart below).

Critical Minerals Policy Positioning



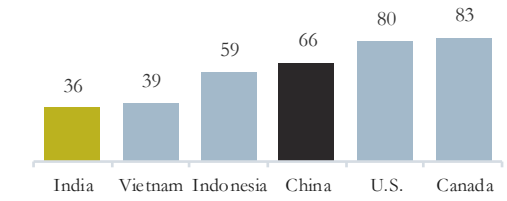
From PwC 2026 Mining Report

China's Share of World Demand (%)



Copper is now ~54% (2024).

Urban Population, % of Total (2025)



China was 36% urban in 2000 and 48% in 2009.

